YAN LUO

Research Fellow | Harvard AI and Robotics Lab

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RESEARCH INTERESTS

My research focuses on multimodal reasoning and responsible AI, with an emphasis on developing trustworthy, fair, and generalizable intelligent systems. Also, I investigate vision-language learning, multimodal large language models, and multi-agent reasoning to enable robust understanding and adaptive reasoning across modalities. My long-term vision is to build intelligent systems that can reason across diverse modalities, adapt to dynamic environments, and make reliable, equitable decisions in complex real-world scenarios.

ACADEMIC APPOINTMENTS

Aug. 2022 - Present	**Harvard AI and Robotics Lab** *Research Fellow* (*Postdoctoral Advisor: Prof. Mengyu Wang & Prof. Tobias Elze*)	MA, USA
Dec. 2024 - Present	**Broad Institute of MIT and Harvard** *Affiliate Postdoctoral Research Fellow* (*Postdoctoral Advisor: Prof. Mengyu Wang*)	MA, USA
Jan. 2017 - Jun. 2022	**University of Minnesota** *Research Assistant* (*Doctoral Advisor: Prof. Qi Zhao*)	MN, USA
May. 2013 - Dec. 2017	**National University of Singapore** *Research Assistant* (*Supervisors: Prof. Mohan Kankanhalli & Prof. Qi Zhao*)	Singapore, Singapore

EDUCATION

Jan. 2017 - Jul. 2022	**University of Minnesota Twin Cities** *Ph.D. in Computer Science* Doctoral Advisor: Prof. Qi Zhao	Minnesota, USA
Jul. 2015 - Jan. 2017	**National University of Singapore** *Ph.D. Student in Computer Engineering* Transferred to University of Minnesota Doctoral Advisor: Prof. Qi Zhao	Singapore, Singapore
Sep. 2004 - Jul. 2008	**Xi'an University of Science and Technology** *B.S. in Computer Science*	Shaanxi, China

HONORS & AWARDS

2023	**Travel Grant** — The Association for Research in Vision and Ophthalmology New Orleans, USA
2007	**2nd Prize in Province** — China Undergraduate Mathematical Contest in Modeling Shaanxi, China
2006	**2nd Prize in Province** — China Undergraduate Mathematical Contest in Modeling Shaanxi, China

SELECTED PUBLICATIONS

Revision-Aware Independent Agent Graphs for Dynamic Reasoning

Yan Luo, Selim-Antoine Lali, Jeremy Moebel, Iliass Khoutaibi, Ahmadou Aidara, and Mengyu Wang

arXiv preprint arXiv:2610.01249 (2026).

Representation-guided in-context learning for medical image interpretation with multimodal large language models

Minda Zhao, Fangyu Hu, **Yan Luo**, Yutong Yang, Jiahui Cai, Kaichen Zhou, Manling Li, Paul Liang, Yilun Du, Lucy Q. Shen et al

arXiv preprint arXiv:2609.24057 (2026).

Foveation-based mechanisms alleviate adversarial examples

Yan Luo, Xavier Boix, Gemma Roig, Tomaso Poggio, and Qi Zhao

arXiv preprint arXiv:1511.06292 (2015).

Fairclip: Harnessing fairness in vision-language learning

Yan Luo, Min Shi, Muhammad Osama Khan, Muhammad Muneeb Afzal, Hao Huang, Shuaihang Yuan, Yu Tian et al

In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 12289-12301. 2024.

FairSeg: A Large-Scale Medical Image Segmentation Dataset for Fairness Learning Using Segment Anything Model with Fair Error-Bound Scaling

Yu Tian, Min Shi, **Yan Luo**, Ava Kouhana, Tobias Elze, and Mengyu Wang

In The Twelfth International Conference on Learning Representations.

Harvard glaucoma detection and progression: A multimodal multitask dataset and generalization-reinforced semi-supervised learning

Yan Luo, Min Shi, Yu Tian, Tobias Elze, and Mengyu Wang

In Proceedings of the IEEE/CVF International Conference on Computer Vision, pp. 20471-20482. 2023.

Learning to predict trustworthiness with steep slope loss

Yan Luo, Yongkang Wong, Mohan S. Kankanhalli, and Qi Zhao

Advances in Neural Information Processing Systems 34 (2021): 21533-21544.

FairDiffusion: Enhancing equity in latent diffusion models via fair Bayesian perturbation

Yan Luo, Muhammad Osama Khan, Congcong Wen, Muhammad Muneeb Afzal, Titus Fidelis Wuermeling, Min Shi, Yu Tian, Yi Fang, and Mengyu Wang

Science Advances 11, no. 14 (2025): eads4593.

RESEARCH & ENGINEERING EXPERIENCE

Aug. 2022 - Present **Harvard Ophthalmology AI Lab** MA, USA
Postdoctoral Research Fellow

- Design and develop novel vision–language deep learning models to bridge human visual cognition and computer vision through multimodal representation learning.
- Apply R statistical methods to medical data to study eye diseases.

Jan. 2017 - Jul. 2022 **Department of Computer Science and Engineering at UMN** MN, USA
Research/Teaching Assistant (*Advisor: Prof. Qi Zhao*)

- Studied how to improve the generalization and transferability of the learning process, and how to assess the trustworthiness of predictions for the safe and effective use of AI models.
- Assisted the instructors in lecturing the undergraduate- and graduate-level courses.

Dec. 2018 - Jul. 2022 **CoTwins Lab at UMN** MN, USA
Research Assistant (*Supervisor: Prof. Scott Vrieze and Prof. Qi Zhao*)

- Developed and maintained iPhone App (CoTwins in the App Store), as well as the server hosted in Google Cloud Project.
- Served as a database administrator for the project. We used the MySQL database that was run in GCP.
- Performed statistical analyses using R.

Jun. 2021 - Aug. 2021 **JD AI Research** CA, USA
Research Intern

- Studied how domain change in the training data affects the continual learning process.
- Devised and developed an algorithm to alleviate catastrophic forgetting that results from the domain change in the training data.

Mar. 2013 - Jan. 2017 **Interactive & Digital Media Institute at NUS** Singapore, Singapore
Research Assistant (*Supervisor: Prof. Mohan Kankanhalli and Prof. Qi Zhao*)

- Proposed and developed a novel and general method to predict important content in visual scenes from multiple sources (e.g., videos/images).
- Designed and conducted human experiments to see how people allocate their attention to multiple sources, and compare results with those of one source as in the conventional experimental setting.
- Developed algorithms to extract features and compute key eye parameters for clinical applications.

Nov. 2010 - Mar. 2013 **SmartClouder Inc.** Beijing, China
R&D Engineer

- Developed a decentralized distributed system that performed better than the NAS product of NetApp Inc. This project addressed the shortcomings (at that time) of the existing commercial NAS system including disruption of the client-side service to reconfigure when the storage infrastructure was upgraded or expanded, the expensive price of commercial NAS product (NetApp FAS3240 costs 450,000 RMB) and the remarkable performance decline when increasing the number of files.
- Built a virtualization system including all functionalities of Amazon EC2 based on KVM virtualization infrastructure and improved its security by applying the VLan technique.

Jul. 2008 - Nov. 2010 **TeleNav Inc.** Beijing, China
R&D Engineer

- Developed algorithms to improve two aspects of traditional route planning engines - increased the performance in long-distance routing (e.g., from Los Angeles to New York) and improved the quality in some particular road networks (e.g., two highways that are parallel at the beginning but intersect at the end). Our work outperforms other commercial geospatial engines in computing at that time.
- Developed a navigation system for mobile device users who cannot connect to a cell phone network (e.g., GSM or CDMA) or where the location is not covered by the network. By effectively addressing the memory limitation, this work was a pioneering work in navigation with mobile devices (where computational resources are highly constrained at that time).

TEACHING

2023 – 2026 **Harvard Medical School** Massachusetts, USA
Co-Lecturer

Artificial Intelligence in Medicine (funded by NIH P30 EY003790)
Spring 2023; Fall 2023; Spring 2024; Spring 2025; Fall 2025; Spring 2026

2018 – 2022 **University of Minnesota** Minnesota, USA
Teaching Assistant

- CSCI 4511W: Introduction to Artificial Intelligence — Spring 2018, Fall 2018, Spring 2019
- CSCI 5481: Computational Techniques for Genomics — Fall 2019
- CSCI 5801: Software Engineering I — Spring 2020, Spring 2021, Spring 2022
- CSCI 4707: Practice of Database Systems — Fall 2021

ACADEMIC SERVICE

Reviewer	IEEE TPAMI, IEEE TNNLS, IEEE TMI, CVPR 2019, ICCV 2019, AAAI 2020, CVPR 2020, ECCV 2020, NeurIPS 2020, WACV 2021, AAAI 2021, CVPR 2021, ICML 2021, ICCV 2021, NeurIPS 2021, WACV 2022, AAAI 2022, CVPR 2022, ICLR 2022, ICML 2022, ECCV 2022, NeurIPS 2022, ACCV 2022, WACV 2023, ICLR 2023, CVPR 2023, ICCV 2023, NeurIPS 2023, ICLR 2024, CVPR 2024, ICML 2024, ECCV 2024, NeurIPS 2024, ICLR 2025, ICML 2025, ICCV 2025, NeurIPS 2025, AAAI 2025, ICLR 2026, CVPR 2026, ICML 2026, ECCV 2026, ICLR 2027, CVPR 2027
Guest Editor	MDPI Sensors Special Issue: Biomedical Sensing and Bioinformatics Processing
Moderator	Association for Research in Vision and Ophthalmology (ARVO) 2024: Poster Session of Computational Modeling, Machine Learning, and Artificial Intelligence Approaches in Glaucoma

SKILLS

AI & Machine Learning	Multimodal Reasoning, Vision-Language Learning, Large Language Models, Multi-Agent Reasoning, Responsible AI, Computer Vision, Medical AI
LLM Tools	Hugging Face, Transformers, PEFT/LoRA, vLLM
Generative AI Platforms	GPT, Claude, Gemini
Programming	Python, R, MATLAB, C/C++, Java, Lua
ML Frameworks	PyTorch, TensorFlow, scikit-learn; Torch, Caffe
Data & Databases	pandas, PySpark, MySQL, Cassandra
Cloud & Computing	Google Cloud Platform, Amazon Web Services, Slurm, PBS Pro
Engineering Tools	Git, Docker, Podman, Linux Shell Scripting
Operating Systems	Linux, macOS, Windows